# HARVEST

## an Agricultural Regression Model based on Historical and Regional Variables for Estimating System Resilience Trends

Presented for INFEST XII Data Science








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# Problem Understanding

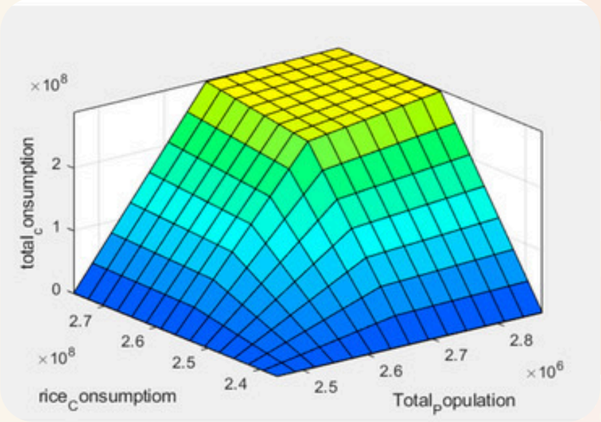
# Problem Statement



Climate change and supply chain disruptions increasingly threaten food system resilience



Food system resilience varies across regions and changes over time




Accurate prediction is needed to anticipate regional food vulnerabilities

A **spatio-temporal predictive approach** is needed to accurately estimate the **2026 Food System Resilience Index**

# Dataset at a Glance

## Data Structure

*Input (X)*

| | region_id | year | avg_temp | total_precip | crop_production | food_imports | gdp_per_capita | ... |
|---|---|---|---|---|---|---|---|---|
| Type | object | int64 | float64 | float64 | float64 | float64 | float64 | ... |
| | IDN_001 | 2010 | 27.15 | 2210.30 | 1523.40 | 342.10 | 3160.50 | ... |
| | IDN_001 | 2011 | 27.32 | 1894.12 | 1480.10 | 365.20 | 3342.80 | ... |
| | IDN_001 | 2012 | 27.48 | 2051.76 | 1601.33 | 389.50 | 3560.10 | ... |
| | IDN_002 | 2010 | 25.61 | 3120.45 | 980.21 | 210.35 | 2280.40 | ... |
| | IDN_002 | 2011 | 25.44 | 2987.33 | 1012.67 | 205.18 | 2401.90 | ... |
| | ... | ... | ... | ... | ... | ... | ... | ... |

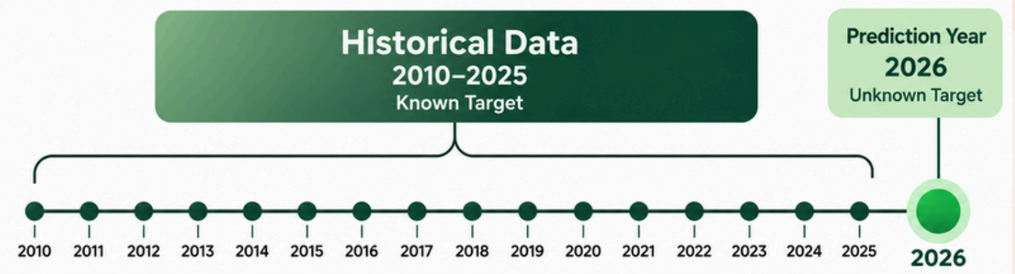
Multi-domain regional predictors

*Target (y)*

| region_id | food_system_resilience_index |
|---|---|
| IND_001 | 37.258407 |
| IND_002 | 39.039688 |
| IND_003 | 48.547394 |
| IND_004 | 54.413943 |
| IND_005 | 44.727757 |
| IND_006 | 44.450290 |
| ... | ... |

Food System Resilience Index

## Dataset Coverage




**358 regional units**

## Performance Metric

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}$$

*Lower is Better*

## Main Challenges

1. **Spatio-Temporal Dependencies**
2. **Limited Information for 2026 Prediction**

# Exploratory Data Analysis

# Data Coverage & Missingness

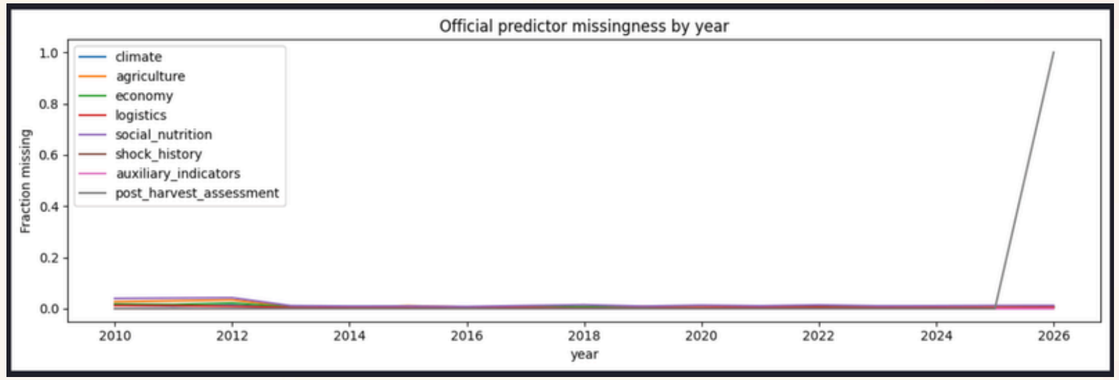
| year | count | mean | std | min | max |
|---|---|---|---|---|---|
| 2010 | 358 | 48.862047 | 14.579193 | 1.463 | 96.619 |
| 2011 | 358 | 48.940877 | 14.606394 | 3.637 | 90.876 |
| 2012 | 358 | 48.332732 | 15.482416 | 3.789 | 99.551 |
| 2013 | 358 | 48.447682 | 15.190410 | 2.287 | 100.000 |
| 2014 | 358 | 48.686480 | 14.991606 | 9.983 | 100.000 |
| 2015 | 358 | 48.267779 | 14.802401 | 6.474 | 99.918 |
| 2016 | 358 | 48.467109 | 14.498275 | 10.649 | 86.951 |
| 2017 | 358 | 48.227327 | 15.187387 | 8.653 | 88.736 |
| 2018 | 358 | 48.275444 | 14.371644 | 11.109 | 89.907 |
| 2019 | 358 | 48.361606 | 15.267179 | 7.172 | 97.456 |
| 2020 | 358 | 47.755592 | 15.432074 | 0.000 | 92.528 |
| 2021 | 358 | 47.992025 | 14.931870 | 1.853 | 84.726 |
| 2022 | 358 | 47.921098 | 14.798943 | 0.000 | 92.750 |
| 2023 | 358 | 47.724679 | 14.975199 | 2.389 | 87.965 |
| 2024 | 358 | 47.689707 | 15.218995 | 6.490 | 91.436 |
| 2025 | 358 | 48.288645 | 14.778504 | 8.778 | 97.385 |

**358 / 358 regions available every year**

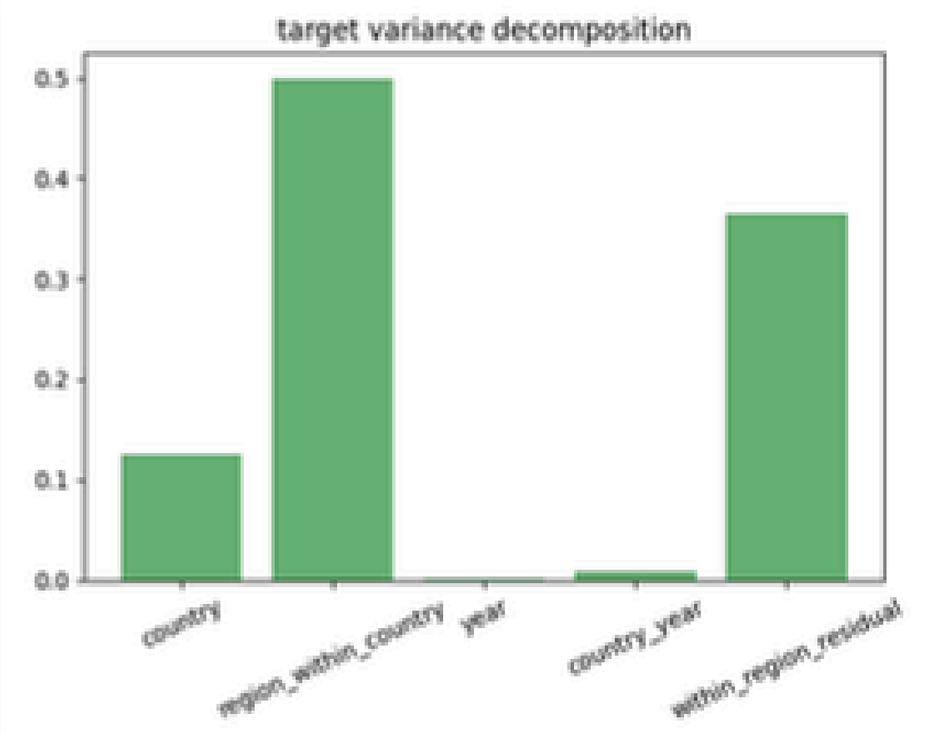
| | table | historical_missing_pct | missing_2026_pct |
|---|---|---|---|
| 0 | climate | 0.545566 | 0.465549 |
| 1 | agriculture | 1.191515 | 0.628492 |
| 2 | economy | 0.739787 | 0.523743 |
| 3 | logistics | 0.535382 | 0.651769 |
| 4 | social_nutrition | 1.699255 | 1.241465 |
| 5 | shock_history | 0.000000 | 0.000000 |
| 6 | auxiliary_indicators | 0.000000 | 0.000000 |
| 7 | post_harvest_assessment | 0.000000 | 100.000000 |

**Historical missingness is relatively low**

***2026 post harvest assessment: 100% missing***



# Regional Structure Domains




**Almost half of the target variance is explained by persistent differences between regions within the same country.**



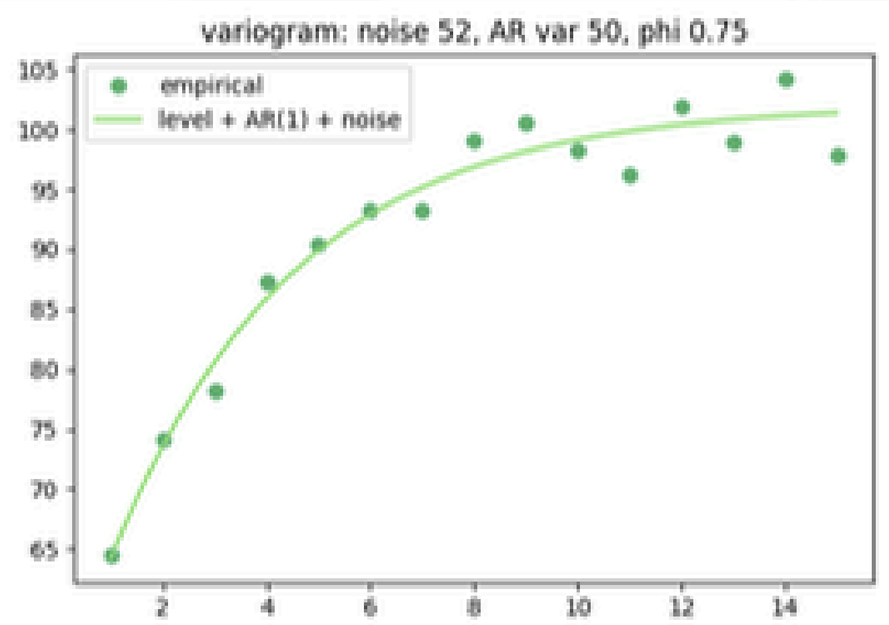
**We need a model based mainly on global temporal trends**

Preserve:

- Regional identity
- Historical regional levels

Food-system resilience is primarily **region-structured,** rather than driven by a universal annual trend.

# Resilience Is Persistent but Noisy Through Time

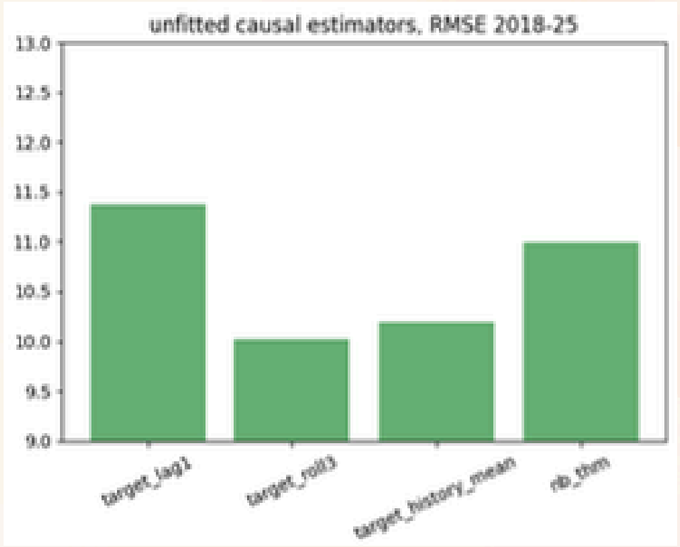



- AR(1) persistence: **Φ =0.754**
- Noise variance **≈ 51.9**
- AR variance **≈ 50.2**

A region's resilience this year strongly informs its future resilience, but **annual fluctuations remain substantial.**

**cell 071 output 01**

| | year | lag1_rmse |
|---|---|---|
| 0 | 2018 | 11.693969 |
| 1 | 2019 | 11.068143 |
| 2 | 2020 | 11.713873 |
| 3 | 2021 | 11.669722 |
| 4 | 2022 | 10.410077 |
| 5 | 2023 | 10.674453 |
| 6 | 2024 | 12.015629 |
| 7 | 2025 | 11.613265 |

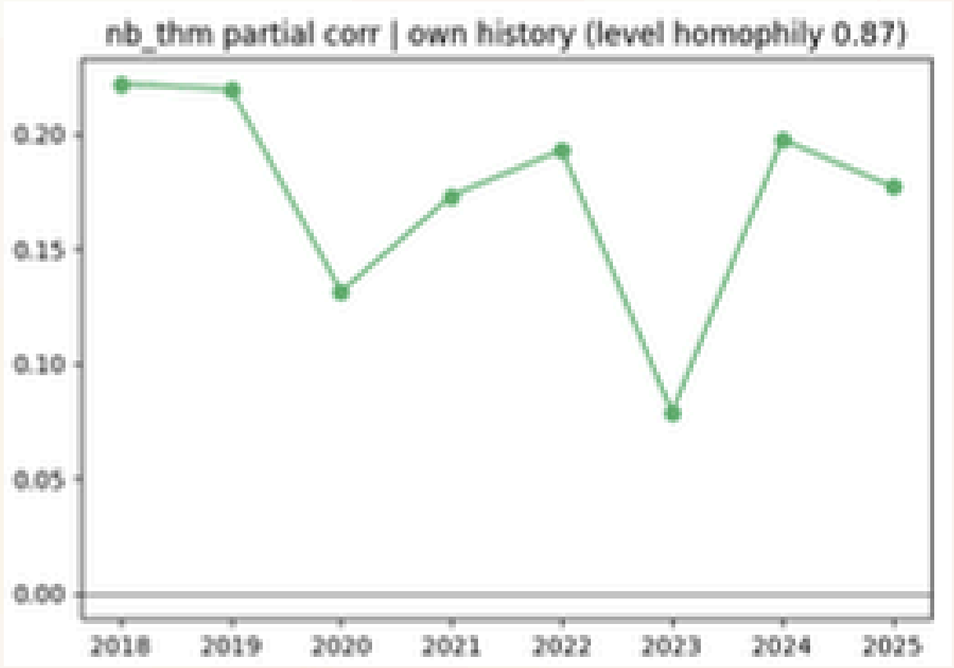



- one year lag is too noisy
- 3-year historical average performs better

persistent multi-year history

Resilience exhibits meaningful temporal memory. **Rolling historical context** is more reliable than a single previous-year observation.

# Spatial Neighbors Carry Additional Signal




Graph-level Homophily **0.870**

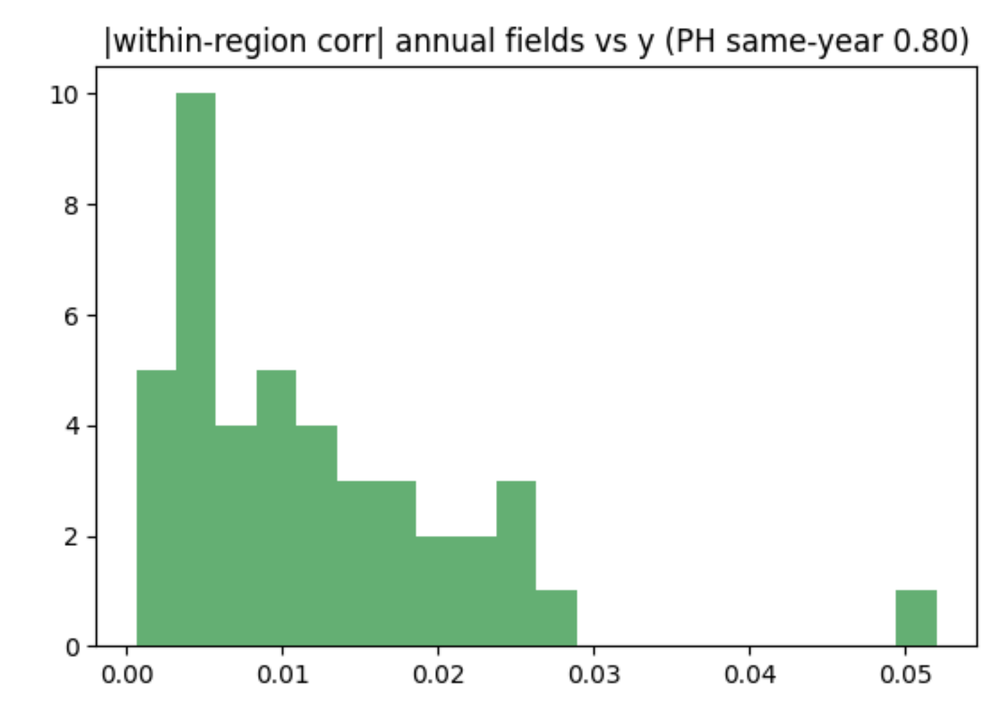
**The graph is not simply duplicating each region's own history,
neighbor information continues to contain additional information.**

| | feature | source | count_2026 |
|---|---|---|---|
| 0 | nb_thm | neighbor target histories through t-1 | 358 |

**nb_thm = weighted mean of neighbors' historical target means**

**Regional resilience is spatially correlated**
Historical neighbor information adds signal beyond a region's own past.

# Post-Harvest Paradox




Most annual predictors have weak individual correlations with resilience, while **same-year post-harvest assessment shows a strong correlation (0.803)**.

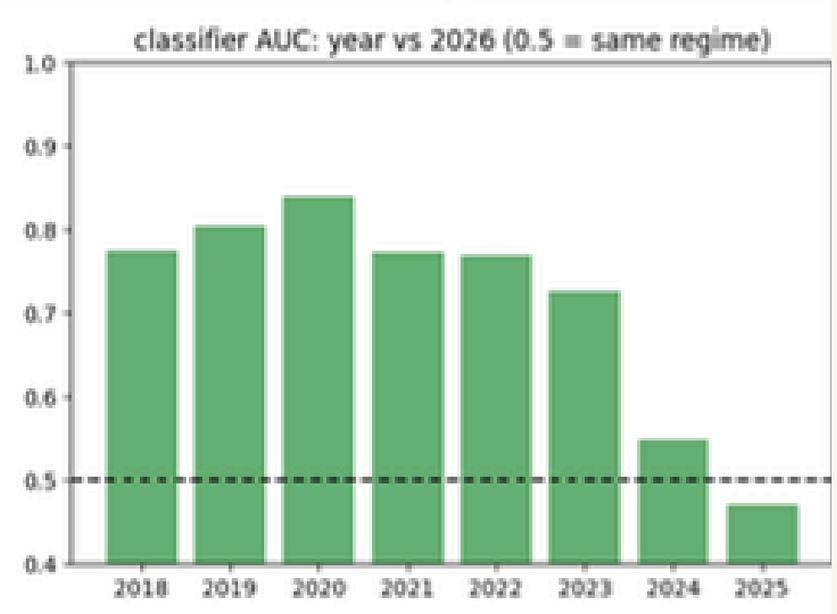
| Same-year PH Correlation | Lagged PH Correlation | PH Missing in 2026 |
|---|---|---|
| **0.803** | **0.176** | **100%** |

**Modelling Implications**

- Exclude actual current-year post-harvest assessment
- Preserve historical features
- Investigate whether a forward-trained proxy can restore useful information.

**The strongest same-year predictor is unavailable in 2026, requiring an alternative forecasting strategy.**

# Historical-to-2026 Regime Shift




Most current-year variables have **very weak individual relationships with the Food System Resilience Index,** even after accounting for regional differences.

**Classification AUC**
2024 vs 2026
**0.549**

**Classification AUC**
2025 vs 2026
**0.470**

Focus model evaluation on recent years, especially 2024–2025, because they are more representative of 2026.

**Recent years provide a more relevant reference for predicting 2026 than older historical years.**

# Preprocessing

# Leakage-Safe Spatio-Temporal Dataset Construction

8 Annual Datasets

Regional Metadata

Neighbor Graph

merge using region x year

Validate Structure

Construct Causal History

Freeze feature contract

Supervised region-year features for 2013–2025 + 2026 inference

# Temporal Feature Engineering




## Temporal Features

Lag Features

$$y_{r,t-1}, y_{r,t-2}$$

Rolling Features

$$target_roll3_{r,t} = \frac{y_{r,t-1} + y_{r,t-2} + y_{r,t-3}}{3}$$

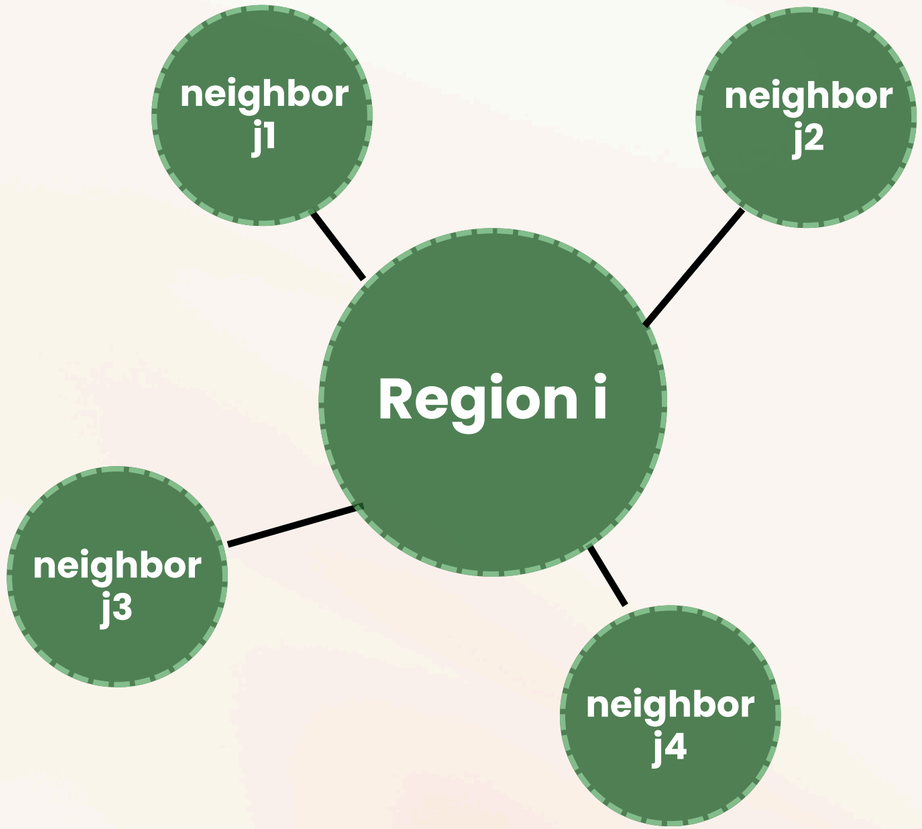
Historical mean

$$\text{Mean}(y_{r,<t})$$

| | feature | causal source | status |
|---|---|---|---|
| 0 | target_lag1/lag2/roll2/roll3/history_mean/delta1 | labels <= t-1 | KEEP (frozen) |
| 1 | ph_lag1/roll3/roll5/history_mean/delta | PH <= t-1 | KEEP (frozen) |
| 2 | nb_thm | official graph, neighbours target_history_mean... | KEEP (frozen) |
| 3 | nb_thm_local / gap x vol / shock recency / res... | - | NOT CREATED |

# Spatial Feature Engineering

## Region + Neighbor Graph




## Historical Neighbor Mean

$$nb_thm_{i,t} = \frac{\sum_{j \in N(i)} w_{ij}\, \bar{y}_{j,<t}}{\sum_{j \in N(i)} w_{ij}}$$

- $i$: region being predicted.
- $t$: prediction year.
- $N(i)$: set of neighboring regions connected to region $i$.
- $w_{ij}$: official neighbor weight between regions $i$ and $j$.
- $\bar{y}_{j,<t}$: average historical resilience index of neighbor $j$ using only years before $t$.
- $nb_thm_{i,t}$: weighted average of neighboring regions' historical resilience.

**Neighbor feature availability = 100%**

| | feature | source | count_2026 |
|---|---|---|---|
| 0 | nb_thm | neighbor target histories through t-1 | 358 |

# Preprocessing Policy

Simple, leakage-safe preprocessing tailored for CatBoost. Each rule answers one data risk.

### Missing values

**CatBoost native NaN**

No imputation. Historical gaps are small (**≤ 1.7% per table**) and CatBoost learns where to route NaN at each split.

### Categorical features

**Native categories**

**region_id, country, climate_zone** enter as raw strings: no one-hot and no target encoding.

### Current-year post-harvest

**Excluded**

**100% missing in 2026**, so it never enters the model. Only its past values (≤ t-1) are used.

### Target & neighbour history

**Shift, then aggregate**

Lags, rolling means and **nb_thm** use labels from years **≤ t-1** only, checked by recomputing them independently.

### Feature scaling

**Not applied**

Tree splits are unaffected by monotone rescaling, so **raw official units** are kept.

### Outliers & outputs

**No clipping**

Raw values preserved (**winsorizing was tested and rejected**); predictions are kept exactly as the model outputs them.

**Minimal by design: raw official values, native NaN and categories, and every history built only from years ≤ t-1.**

# Frozen Feature Contract: 61 Features

**61 features** = 43 current-year + 18 history, graph & identity

## Current-year annual indicators — 43

| | |
|---|---|
| **Climate** (7) | rainfall, temperature, drought, floods, heatwave & extreme-rain days, anomaly |
| **Agriculture** (8) | irrigation, fertilizer, mechanization, seed resilience, crop diversity, rice & maize yield, stability |
| **Economy** (8) | GDP per capita, poverty, unemployment, food CPI, fuel price, import, FX & remittance exposure |
| **Logistics** (6) | road density, port & market access, cold storage, disruption, travel time to market |
| **Social & nutrition** (9) | HDI, internet, social protection, stunting, calories, food spending, governance, early warning, health |
| **Shock history** (1) | shock event flag for the year |
| **Auxiliary** (4) | satellite composite, mobile connectivity, survey latency, admin code |

## History, graph & identity — 18

**Target history** (6) — ≤ t-1
lag 1–2, rolling 2–3, history mean, delta

**Post-harvest history** (5) — ≤ t-1
lag 1, rolling 3 & 5, history mean, delta

**Spatial neighbour** (1) — ≤ t-1
nb_thm: weighted mean of neighbours' past targets

**Identity & static** (6)
region_id, country, climate_zone, year, capital flag, coastal exposure

**Frozen and deployment-safe: all 61 columns exist for 2026 same-year post-harvest and target values are never used.**

# Forward Post-Harvest Proxy Experiment

**Train proxy ≤ 2023** → **Predict PH 2024**

**Train proxy ≤ 2024** → **Predict PH 2025**

**Train proxy ≤ 2025** → **Predict PH 2026**

### Separate model for post-harvest prediction

A Catboost model is trained only on post-harvest information before the prediciton year

### Historical performance

Validation RMSE ranges from 11.45 - 12.75

### 2026 prediction

2026 proxy has no PH label, so true error cannot be measured.
Predicted PH value is proposed as a feature.

| | predict_year | proxy_fit_max_year | proxy_train_rows | proxy_rmse |
|---|---|---|---|---|
| 0 | 2013 | 2012 | 1074 | 11.993362 |
| 1 | 2014 | 2013 | 1432 | 11.889691 |
| 2 | 2015 | 2014 | 1790 | 12.439016 |
| 3 | 2016 | 2015 | 2148 | 11.503473 |
| 4 | 2017 | 2016 | 2506 | 12.013322 |
| 5 | 2018 | 2017 | 2864 | 11.729465 |
| 6 | 2019 | 2018 | 3222 | 11.965879 |
| 7 | 2020 | 2019 | 3580 | 12.752471 |
| 8 | 2021 | 2020 | 3938 | 12.023545 |
| 9 | 2022 | 2021 | 4296 | 11.242491 |
| 10 | 2023 | 2022 | 4654 | 11.507109 |
| 11 | 2024 | 2023 | 5012 | 11.841241 |
| 12 | 2025 | 2024 | 5370 | 11.949714 |
| 13 | 2026 | 2025 | 5728 | NaN |

### Key Takeaway

**does approximation improve future resilience forecasts?**

# Modelling

# Model Exploration

Standard learner  EDA-driven hypothesis

**Baseline**

## Persistence

Carry last year forward: $\hat{y} = y(t-1)$. The bar every model must clear.

**11.370** pooled OOF RMSE

**Gradient boosting**

## LightGBM

Leaf-wise GBDT on the same 61 features, no tuning.

**9.556** pooled OOF RMSE

**Gradient boosting**

## XGBoost

Depth-wise GBDT on the same 61 features, no tuning.

**9.541** pooled OOF RMSE

**Randomized trees**

## ExtraTrees

Bagged trees with random splits, aiming to cut prediction variance.

**9.288** pooled OOF RMSE

**Linear shrinkage**

## Empirical-Bayes Ridge

Linear model shrunk toward each region's historical level.

**9.361** pooled OOF RMSE

**State-space**

## Kalman Filter

Level + AR(1) + noise from EDA ($\phi$ = 0.75), with a spatial prior.

**9.363** pooled OOF RMSE

**History only**

## CatBoost $F_H$

Only 18 history, graph and identity features. Do annual indicators help?

**9.333** pooled OOF RMSE

**Proxy feature**

## CatBoost + PH Proxy

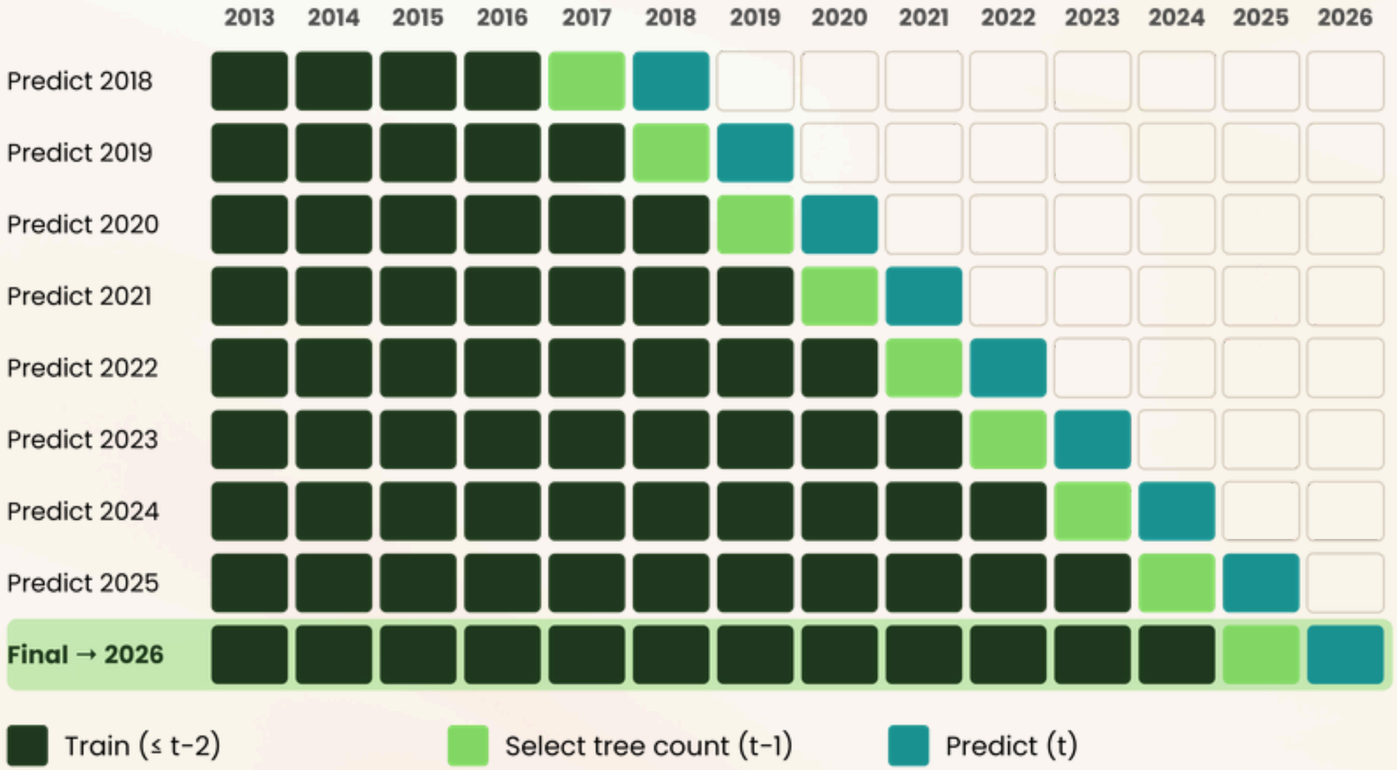
$F_N$ plus the forward post-harvest proxy from slide 18.

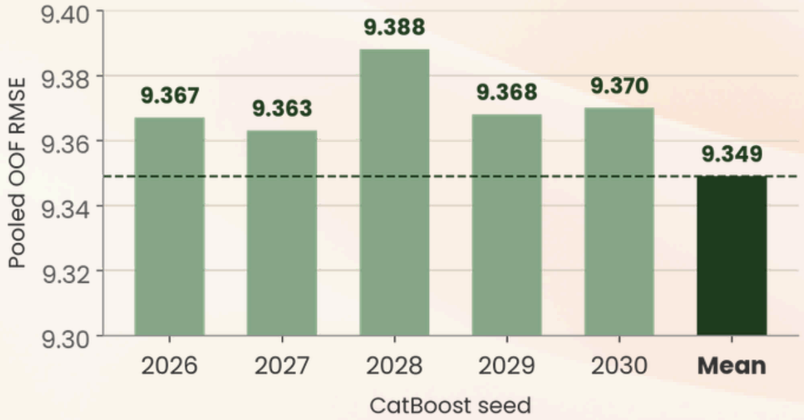
**9.363** pooled OOF RMSE

**Pooled OOF ranks ExtraTrees and a hybrid (9.274) above CatBoost Fn (9.349)**

# Validation Strategy

## Nested Rolling-Origin




Averaging five seeds beats **every** single seed: 9.349 vs 9.363–9.388.

### Evaluation windows

| Pooled | Recent | Deploy |
|---|---|---|
| 2018–2025 | 2023–2025 | 2024–2025 |
| 2,864 rows | 1,074 rows | 716 rows |

*Deploy window: longest training history and closest to 2026 (slide 11).*

**Every score comes from a year the model has never seen.**
**Tree count is chosen one year earlier, never on the year being scored.**

# HARVEST Architecture

**INPUT**
**61 features**
Frozen $F_N$ contract (slide 17)

**LEARNER**
**CatBoost**
Ordered boosting, from scratch

**ENSEMBLE**
**5 seeds**
Seeds 2026–2030, averaged

**OUTPUT**
**2026 index**
358 regions, no manual edits

| Setting | Value |
|---|---|
| Boosting type | Ordered boosting, CPU |
| Depth / learning rate | 6 / 0.03 |
| L2 leaf regularization | 5.0 |
| Tree count | ≤ 2,000, early-stopped (100) on year t-1 |
| Final 2026 fit | Trained on 2013–2025, ≈418 trees (5-seed mean) |
| Categorical features | region_id, country, climate_zone (native) |
| Missing values | Native NaN handling, no imputation |
| Tuning | No HPO; Public LB never used for selection |

## Why CatBoost?

**Ordered boosting**
Unbiased target statistics on a small panel (5,728 labelled rows) limit overfitting to noise.

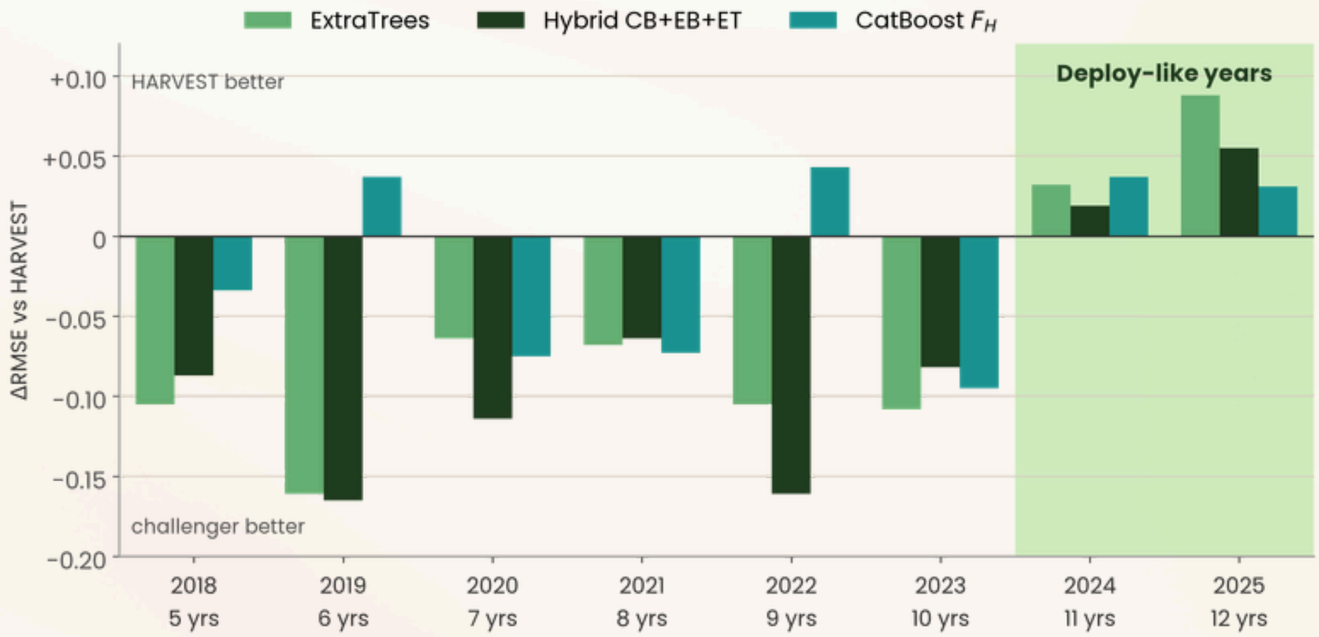
**Native region_id**
Keeps each region's persistent level: ~50% of target variance is region-within-country.

**Native missing values**
No imputation step, so the leakage-safe preprocessing stays intact.

**Best GBDT on equal features**
OOF RMSE 9.349 vs XGBoost 9.541 and LightGBM 9.556.

**One untuned CatBoost, five seeds.**
The accuracy comes from causal features, not model complexity

# Model Selection




Every challenger wins the short-history years but **loses 2024 and 2025,** the folds trained on 11–12 years. The 2026 model is trained on 13.

## Selection Gates

| A | B | C |
|---|---|---|
| Wins 2024–25 | 2026 trend < 0 | Stable over seeds |

| Model | Pooled | 2024–25 | Won | 2026 Δ | Verdict |
|---|---|---|---|---|---|
| ExtraTrees | 9.288 | 9.302 | 6/8 | +0.054 | Reject |
| Hybrid CB+EB+ET | 9.274 | 9.279 | 6/8 | +0.028 | Reject |
| CatBoost $F_H$ | 9.333 | 9.277 | 4/8 | +0.012 | Reject |
| CB + PH proxy | 9.363 | 9.253 | 2/8 | +0.015 | Reject |
| Kalman | 9.363 | 9.450 | 3/8 | +0.175 | Reject |
| **HARVEST** | **9.349** | **9.243** | — | ref. | Select |

*Won = outer years beating HARVEST. 2026 Δ = linear trend of yearly ΔRMSE vs training-history length, extrapolated to 13 years.*

**0/5 Challengers pass all three gates**

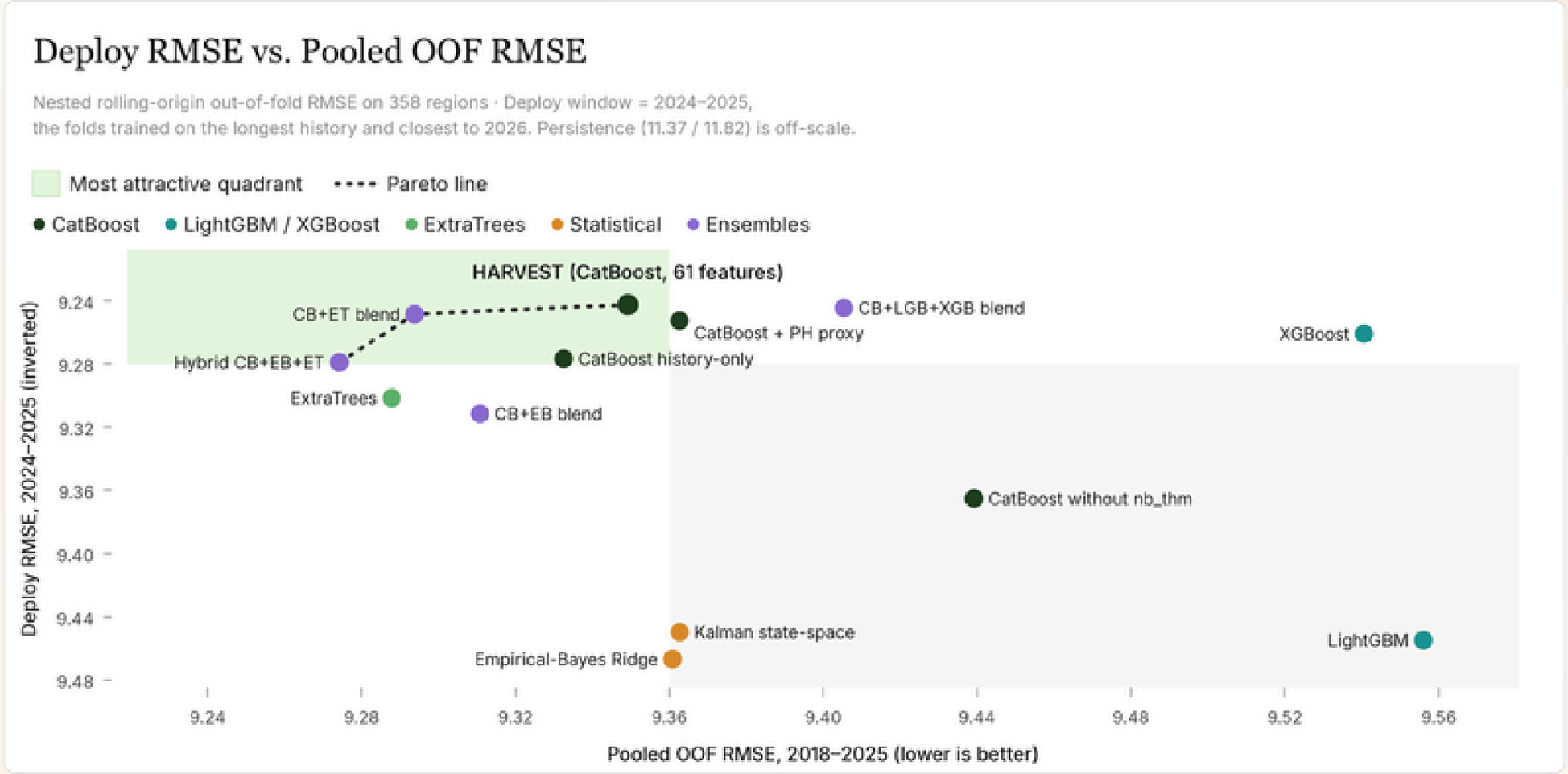
**HARVEST is the only model that wins the years most like 2026.**
**Selected by historical validation, never by the Public LB.**

# Results and Analysis

# Benchmarking

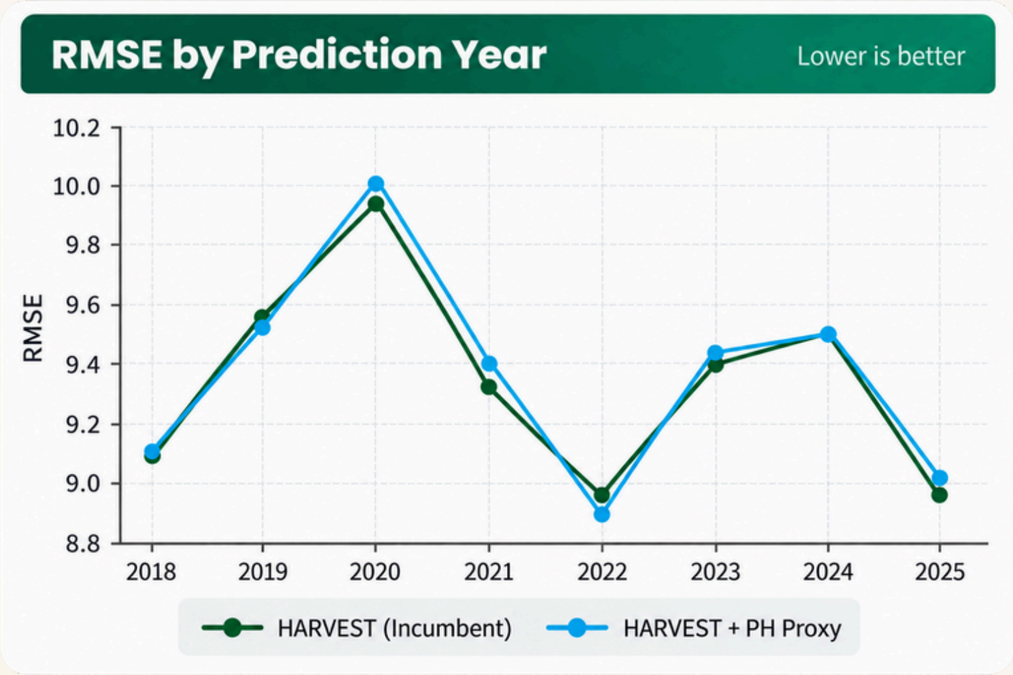
| Model | Feature Configuration | OOF RMSE | 2024–25 RMSE |
|---|---|---|---|
| Persistence | Target lag-1 | 11.370 | 11.816 |
| LightGBM | $F_N$ (61 features) | 9.556 | 9.455 |
| XGBoost | $F_N$ (61 features) | 9.541 | 9.261 |
| CatBoost | $F_0$ (60 features) | 9.446 | 9.380 |
| CatBoost | $F_H$ (18 features) | 9.333 | 9.277 |
| ExtraTrees | $F_N$ (61 features) | 9.288 | 9.302 |
| Hybrid Ensemble | CB + EB + ET | **9.274** | 9.279 |
| CatBoost + PH Proxy | $F_N$ + proxy (62 features) | 9.363 | 9.253 |
| **HARVEST (CatBoost)** | $F_N$ **(61 features)** | **9.349** | **9.243** |

# Benchmarking



# Validation Results

## Yearly RMSE




## Model Selection

### Selection Summary

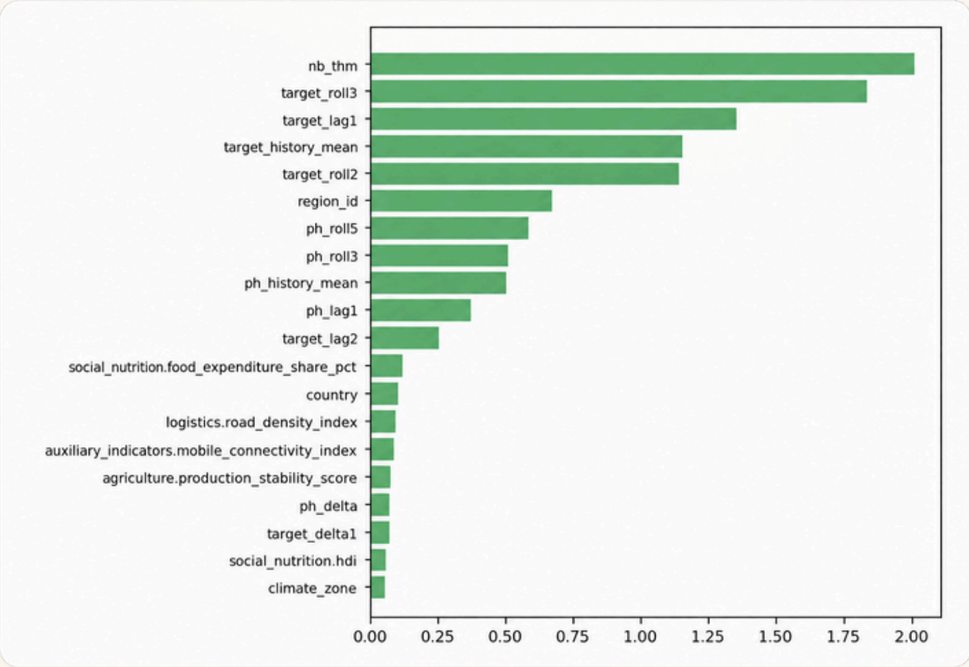
| Model | Pooled OOF RMSE | 2024–25 RMSE | Status |
|---|---|---|---|
| HARVEST (Incumbent) | 9.349 | 9.243 | Selected |
| HARVEST + PH Proxy | 9.363 | 9.253 | Rejected |

1. HARVEST wins in 6 of 8 years
2. Proxy wins only in 2019 and 2022
3. HARVEST outperforms in 2024–2025

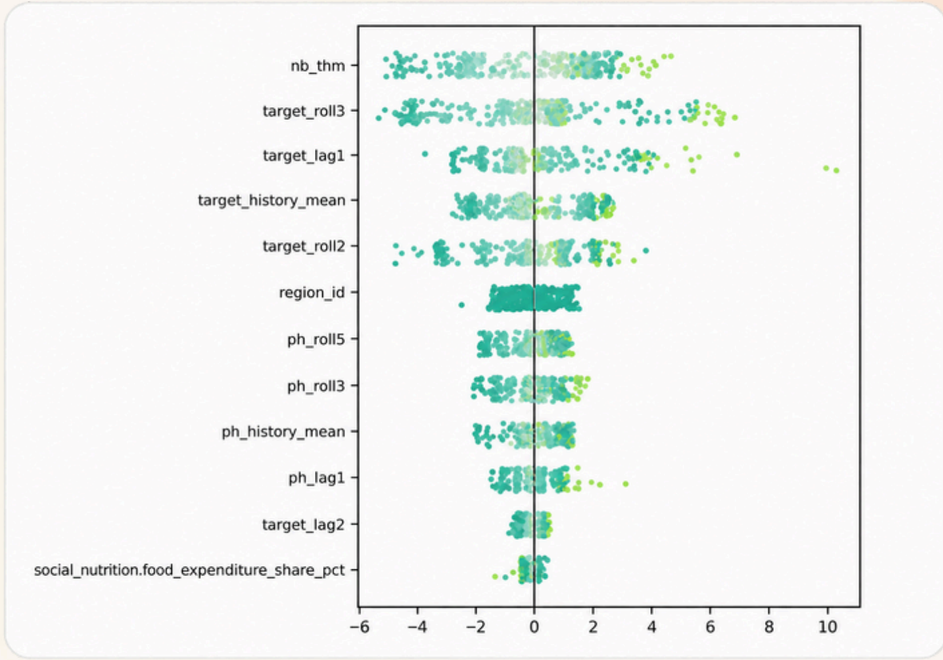
**HARVEST** achieves **lower RMSE** and **more consistent performance** than the post-harvest proxy

# SHAP Analysis

## Feature Importance




## SHAP Distribution




**Top predictor: nb_thm**

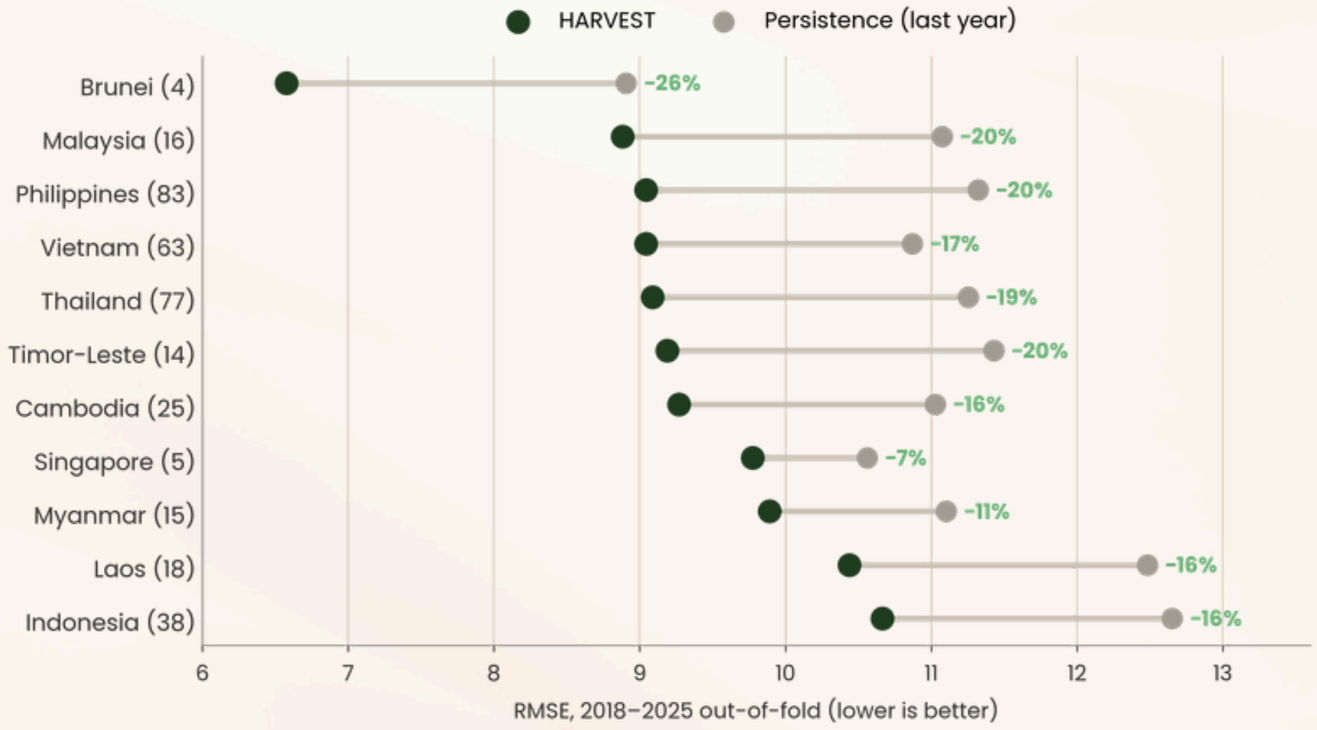
**History + graph + identity: 87.9%**

**Annual fields: 10.3%**

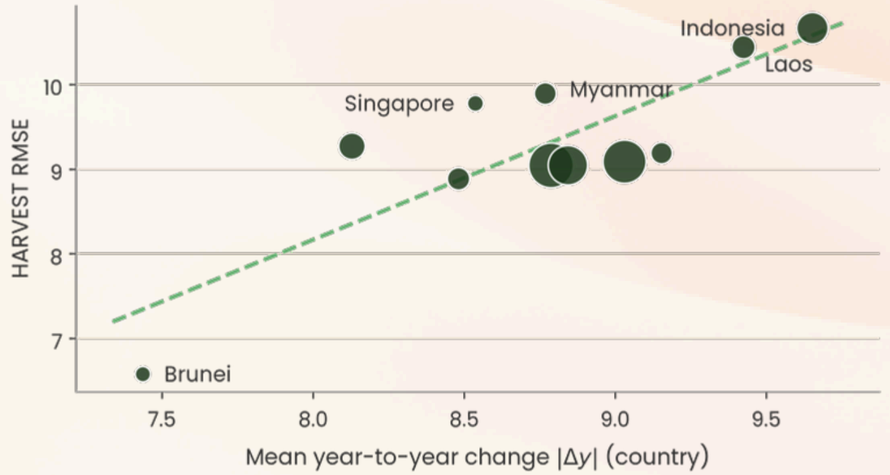
HARVEST predictions are primarily driven by **historical resilience, spatial relationships, and regional identity,** consistent with the **spatio-temporal patterns** identified during EDA.

# Regional Performance

## HARVEST vs Persistence by Country




## Errors Follows Volatility




**Bubble size = regions per country. Dashed line = linear fit**

**11 / 11**

countries where HARVEST beats persistence (-7% to -26% RMSE)

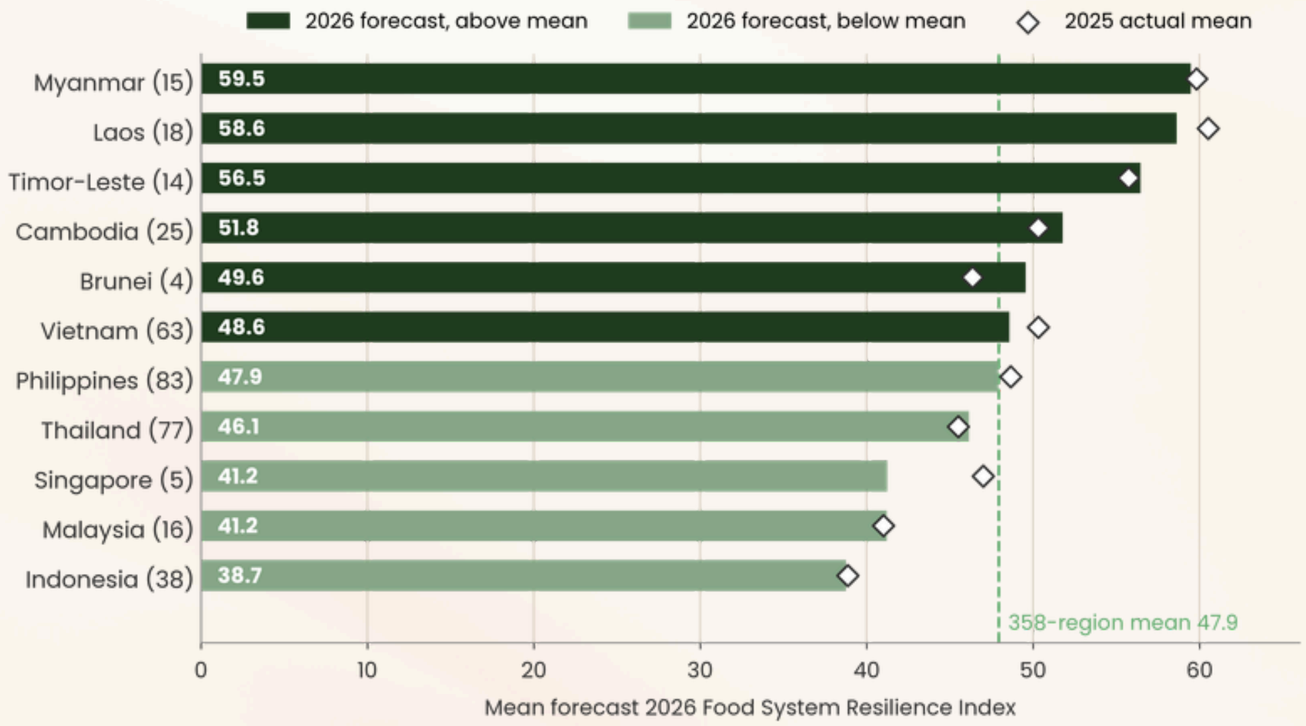
**r = 0.83**

between country error and its natural year-to-year volatility

**HARVEST improves every country. What error remains tracks each country's natural volatility, not model bias (|bias| ≤ 1.6).**

# 2026 Forecast

## Mean 2026 Forecast by Country




**358**
regions forecast for 2026

**47.9**
mean index, SD 11.9

**0.94**
correlation with each region's history mean

### Most resilient in 2026

| Region | Country | 2026 | 2025 actual |
|---|---|---|---|
| **Ilocos Norte** | Philippines | **83.2** | 91.0 |
| **Nueva Vizcaya** | Philippines | **80.6** | 97.4 |
| **Bohol** | Philippines | **74.4** | 82.4 |

### Most vulnerable in 2026

| Region | Country | 2026 | 2025 actual |
|---|---|---|---|
| **Kalimantan Tengah** | Indonesia | **21.1** | 20.7 |
| **Kalimantan Barat** | Indonesia | **22.0** | 18.7 |
| **Melaka** | Malaysia | **22.7** | 27.4 |

*Expected error ≈ 9.2 RMSE (2024–2025 validation).*

**Anchored to long-run regional levels: Indonesia (38.7) stays the most vulnerable, Myanmar (59.5) and Laos (58.6) the most resilient.**

# Summary

INSIGHT

**Built on what actually drives resilience.** Resilience is region-structured, persistent (AR(1) $\phi = 0.75$) and spatially shared (homophily 0.87), so HARVEST learns from **history and neighbours**, not noise.

**~50%**

of variance comes from region identity

RIGOR

**Leak-proof by design.** 61 causal features and nested rolling-origin validation: **every score is earned on a year the model never saw**, and the model was chosen without ever looking at the leaderboard.

**0**

leaked labels, zero Public-LB tuning

RESULT

**The strongest model where it matters.** HARVEST delivers the **lowest 2024–2025 RMSE of all 13 models** and 18% less error than persistence, with just one untuned CatBoost and five seeds.

**#1**

deploy RMSE 9.243 among 13 models tested

# Matur Nuwun!